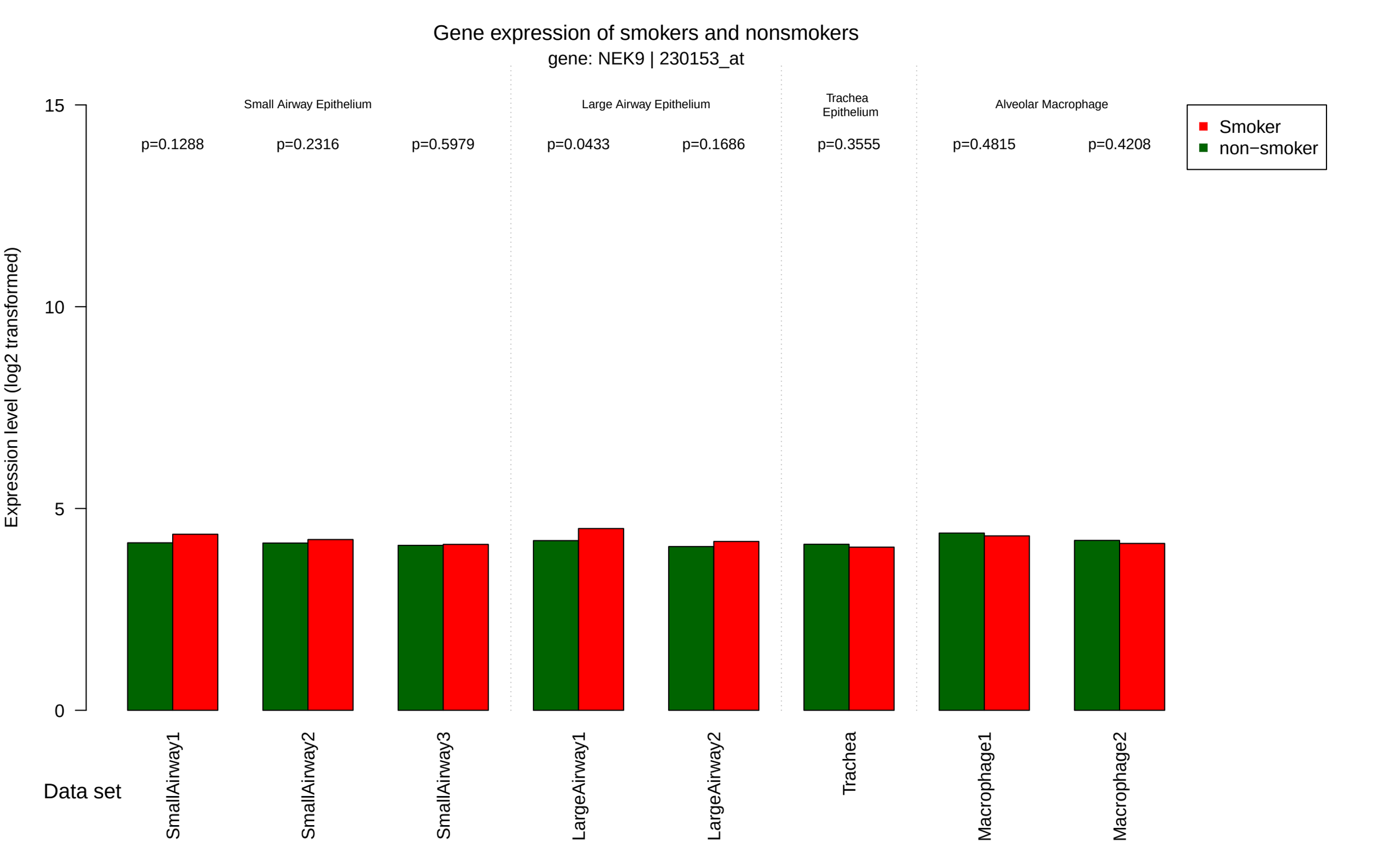
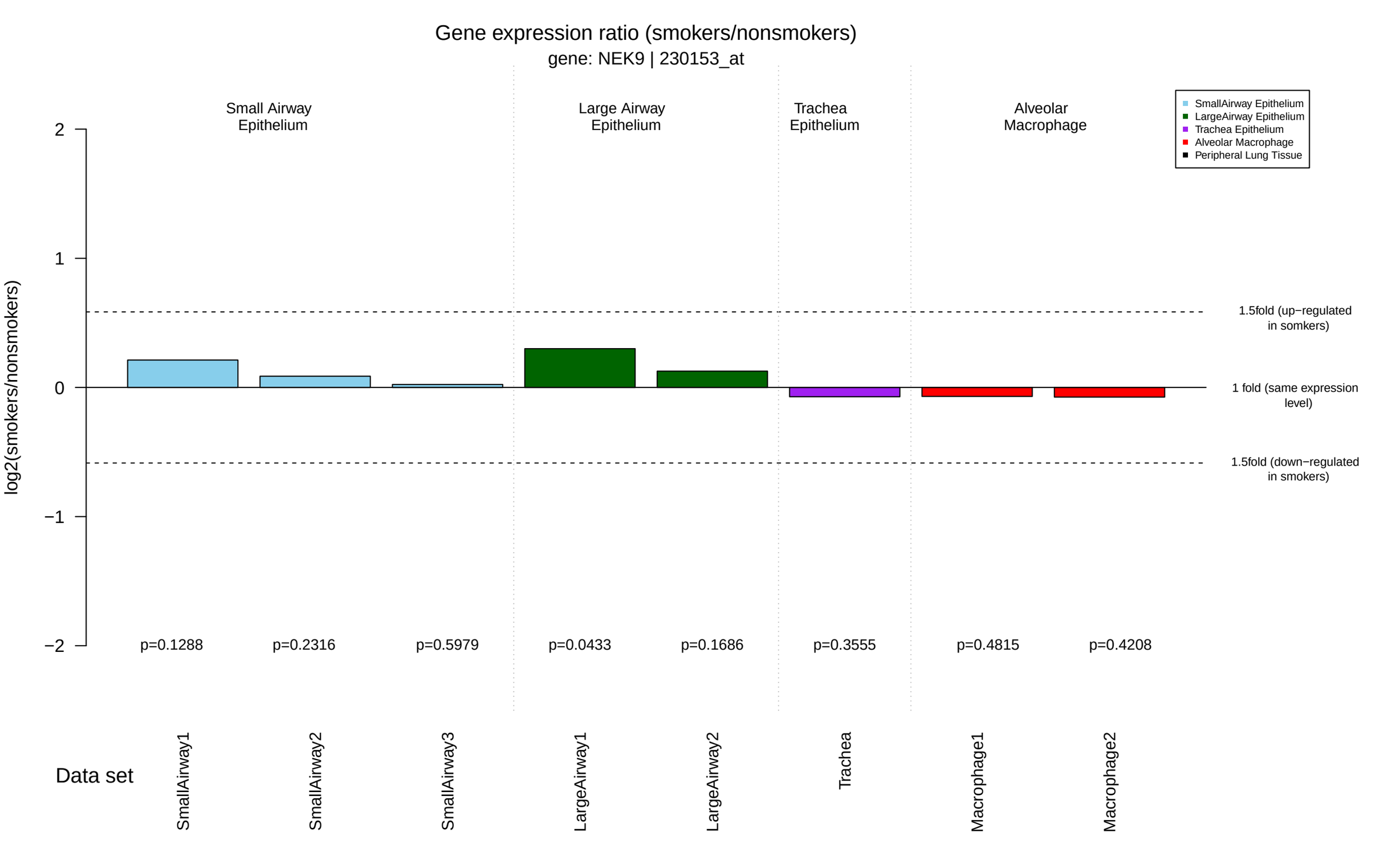
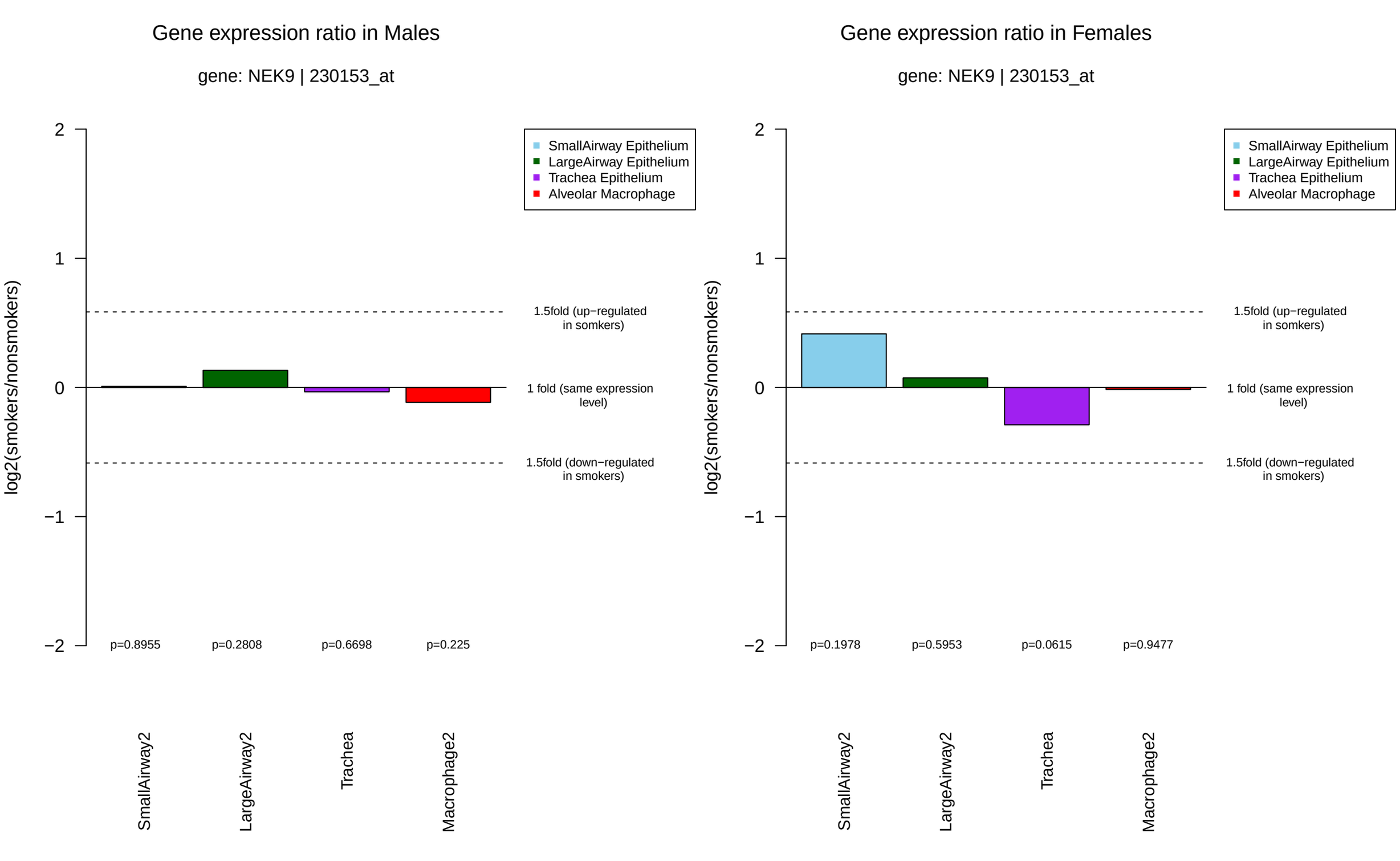
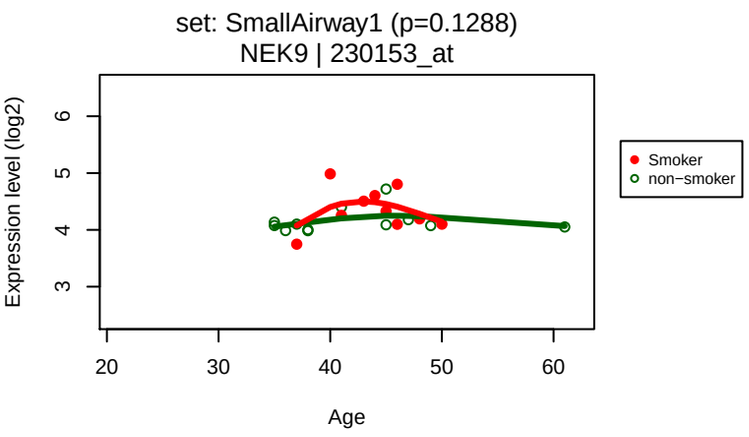
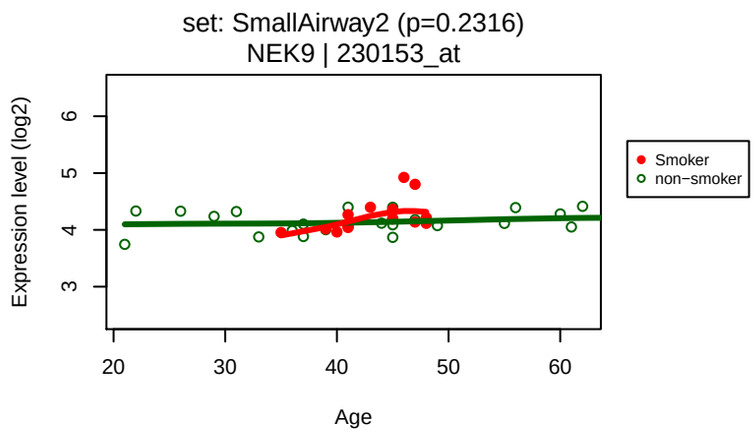
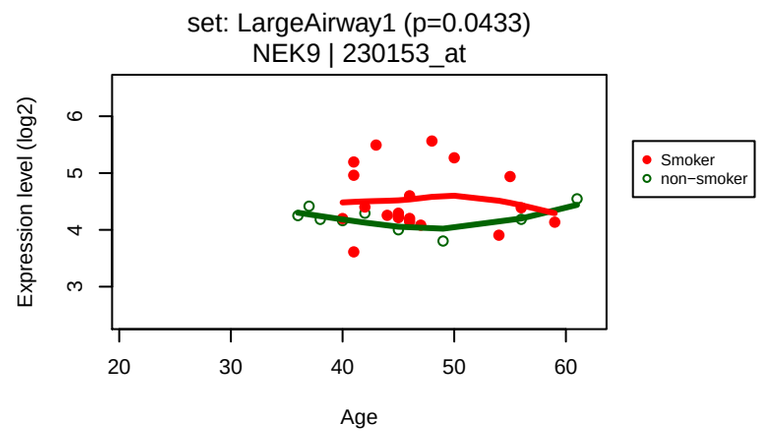

Gene expression of smokers and nonsmokers
gene: NEK9 | 230153_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1288
p=0.2316
p=0.5979
p=0.0433
p=0.1686
p=0.3555
p=0.4815
p=0.4208
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

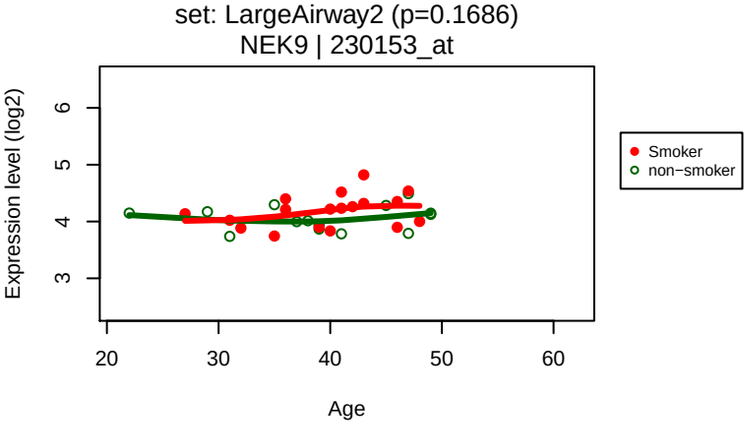

Gene expression ratio (smokers/nonsmokers)
gene: NEK9 | 230153_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1288
p=0.2316
p=0.5979
p=0.0433
p=0.1686
p=0.3555
p=0.4815
p=0.4208
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

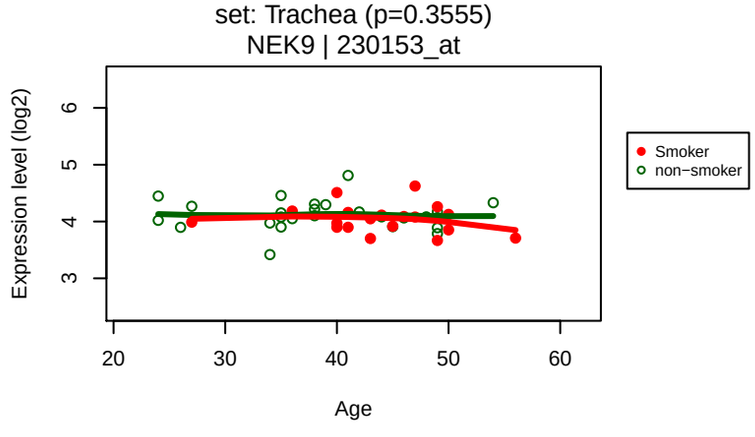

Gene expression ratio in Males
gene: NEK9 | 230153_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8955
p=0.2808
p=0.6698
p=0.225
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NEK9 | 230153_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1978
p=0.5953
p=0.0615
p=0.9477
SmallAirway2
LargeAirway2
Trachea
Macrophage2

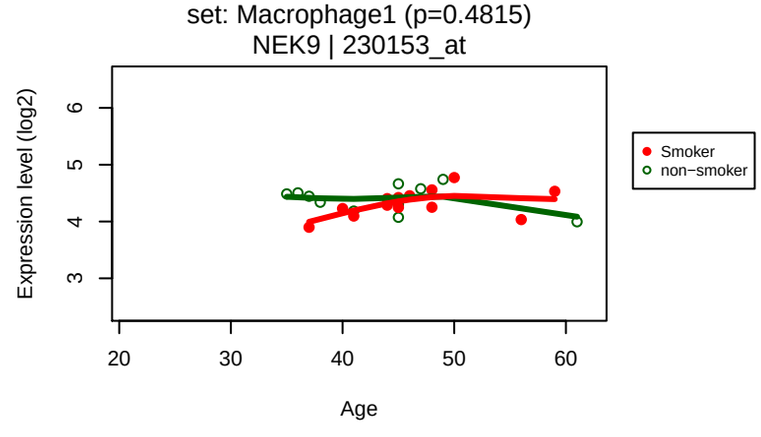

set: SmallAirway1 (p=0.1288)
NEK9 | 230153_at
Expression level (log2)
Age
Smoker
non−smoker

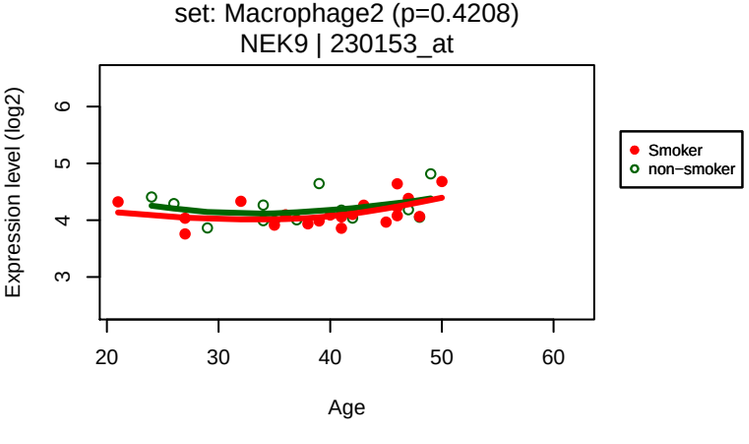

set: SmallAirway2 (p=0.2316)
NEK9 | 230153_at
Expression level (log2)
Age
Smoker
non−smoker

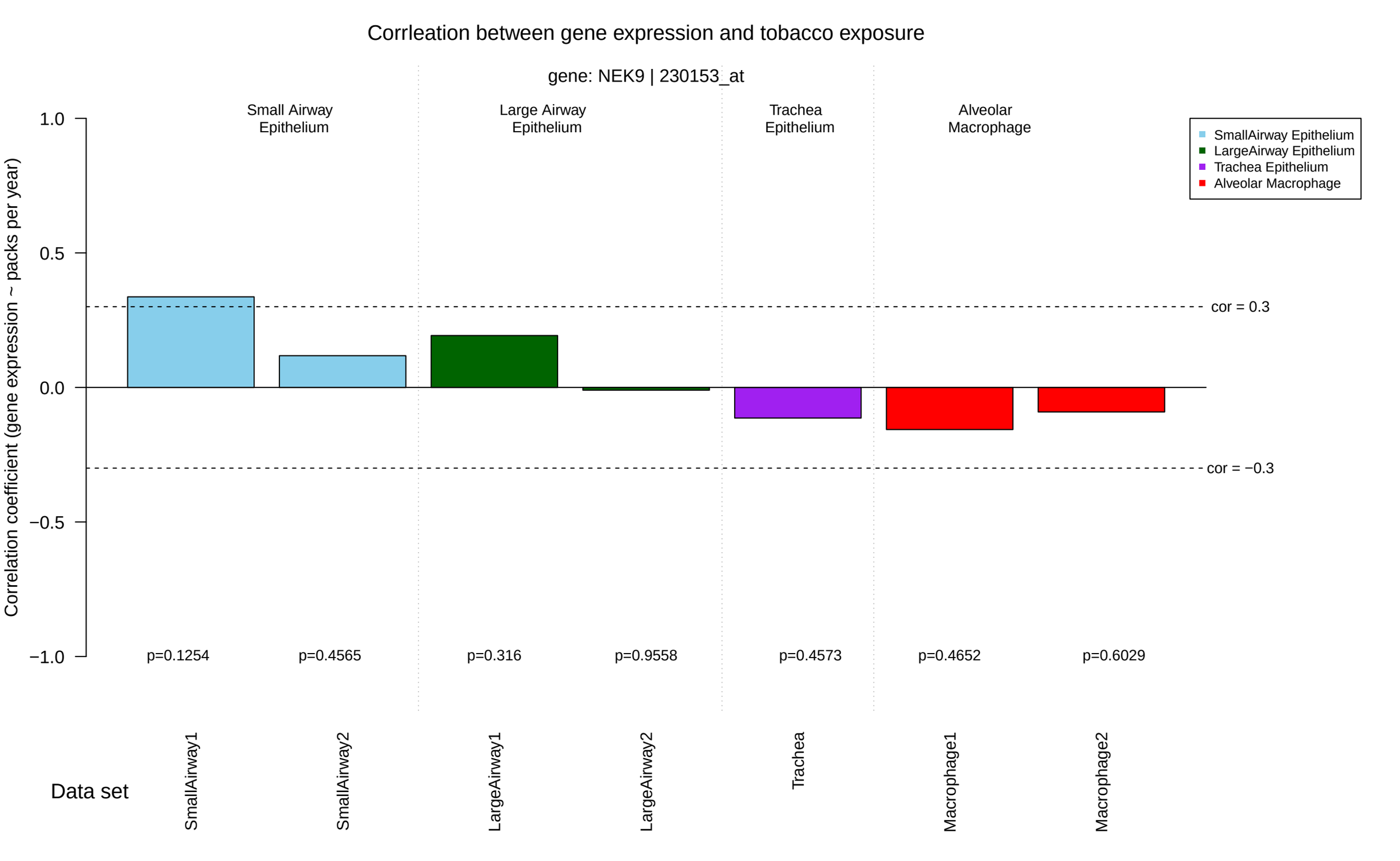

set: LargeAirway1 (p=0.0433)
NEK9 | 230153_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1686)
NEK9 | 230153_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3555)
NEK9 | 230153_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4815)
NEK9 | 230153_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4208)
NEK9 | 230153_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NEK9 | 230153_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1254
p=0.4565
p=0.316
p=0.9558
p=0.4573
p=0.4652
p=0.6029
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2